

```

1  %{
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include "ex4.tab.h"
5  %}
6
7  %%
8
9  [0-9]+ {
10         yyval.num = atoi(yytext);
11         return NUM;
12     }
13
14     [ \t]+ ;
15
16     \n {
17         return EOL;
18     }
19
20     [-+*/()] {
21         return yytext[0];
22     }
23
24     . {
25         fprintf(stderr, "Error: Invalid Character: %s\n", yytext);
26     }
27
28     %%
29
30     int yywrap()
31     {
32         return 1;
33     }
34

```

```

1  %{
2  #include <stdio.h>
3  #include <stdlib.h>
4  #include <string.h>
5
6  int temp_count = 0;
7
8  void yyerror(const char *s)
9  {
10     fprintf(stderr, "Error: %s\n", s);
11 }
12
13 int yylex();
14 %}
15
16 %code requires {
17     typedef struct {
18         int value;
19         char place[20];
20     } Expr;
21 }
22
23 %union {
24     int num;
25     Expr expr;
26 }
27
28 %token <num> NUM
29 %token EOL
30
31 %type <expr> expr
32
33 %left '+' '-'
34 %left '*' '/'
35
36 %%
37
38 program:
39     lines
40     ;
41
42 lines:
43     lines line
44     | line
45     ;

```

```

46
47 line:
48     expr EOL
49     {
50         printf("Result: %s\n", $1.place);
51     }
52 ;
53
54 expr:
55     NUM
56     {
57         $$ = $1;
58         sprintf($$.place, "%d", $1);
59     }
60
61 | '(' expr ')'
62 {
63     $$ = $2;
64 }
65
66 | expr '+' expr
67 {
68     Expr temp;
69     temp.value = $1.value + $3.value;
70     sprintf(temp.place, "%d", ++temp_count);
71
72     printf("%s = %s + %s\n",
73         temp.place, $1.place, $3.place);
74
75     $$ = temp;
76 }
77
78 | expr '-' expr
79 {
80     Expr temp;
81     temp.value = $1.value - $3.value;
82     sprintf(temp.place, "%d", ++temp_count);
83
84     printf("%s = %s - %s\n",
85         temp.place, $1.place, $3.place);
86
87     $$ = temp;
88 }
89
90 | expr '*' expr
91
92 {
93     Expr temp;
94     temp.value = $1.value * $3.value;
95     sprintf(temp.place, "%d", ++temp_count);
96
97     printf("%s = %s * %s\n",
98         temp.place, $1.place, $3.place);
99
100     $$ = temp;
101 }
102 | expr '/' expr
103 {
104     if ($3.value == 0)
105     {
106         yyerror("Division by zero");
107         exit(1);
108     }
109
110     Expr temp;
111     temp.value = $1.value / $3.value;
112     sprintf(temp.place, "%d", ++temp_count);
113
114     printf("%s = %s / %s\n",
115         temp.place, $1.place, $3.place);
116
117     $$ = temp;
118 }
119 ;
120 %%
121
122 int main()
123 {
124     yyparse();
125     return 0;
126 }

```

```

M /c/Users/Afroz/Desktop/EXP7
Afroz@DESKTOP-TDQFQAL UCRT64 /c/Users/Afroz/Desktop/EXP7
$ ./ex4
(12+8)*3-18/(2+1)
t1 = 12 + 8
t2 = t1 * 3
t3 = 2 + 1
t4 = 18 / t3
t5 = t2 - t4
Result: t5

```